UnholyPacketFlow Problem

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Difficulty: Medium-Hard

Time limit per test: 1s

Description

Hitman is a curious and lazy man. He's been hearing all the hype about 5G, edge computing, and blazing-fast internet, but there's just one problem — his download speed still sucks.

Being the genius (and cheapskate) that he is, he's decided to hijack the city's 5G infrastructure and place his own User Plane Function **(UPF)** nodes to maximize speed while minimizing interference.

The city is represented as an **n × n** grid of network zones, numbered from (1,1) in the top-left to (n,n) in the bottom-right. Hitman starts his connection hijack from zone **(1,1)** and wants to route his data stream all the way to **(n,n).** At each point, he can route data right or down to an adjacent zone — any other movement will trigger network security (and he's not ready to fight firewalls today).

Now, each zone **(i, j)** comes with a signal interference cost, calculated based on the zone's alignment with the core infrastructure:

interference(i, j) = gcd(i, a) + gcd(j, b)

Where:

• a and b are the frequency alignment parameters of the vertical and horizontal UPF grid (you can think of them as the secret frequencies he's trying to match).

• **gcd(x, y)** is the greatest common divisor.

Your mission: Help Hitman find the path from **(1,1) to (n,n)** with the minimum total interference cost (yes, he's lazy and doesn't want to deal with too much noise). Be sure to include the costs of the starting and ending zones as well.

Input

The only line contains three integers n,a,b.

Output

Output a single integer: Print one integer —the path from (1,1) to (n,n) with the minimum total interference cost.

Examples

Input	**Output**
2 1 1	6